Advanced BO I: Multi-Objective Bayesian Optimization

MSEN 489/655 - Materials Design Studio

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Outline

1. Single-objective BO recap
2. From one objective to two
3. Key definitions
4. Three-objective case
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Single-objective BO recap

Reminder: how single-objective BO works

Setup

We want to **maximize** an expensive black-box function using as few evaluations as possible:

$$\max_{x \in \mathcal{X}} f(x).$$

- Fit a surrogate model (usually a GP) to the observed data
- Use the GP mean and uncertainty to score candidate points
- Choose the next point by maximizing an acquisition function
- Evaluate the real function there and update the model

Key idea

Do not ask *"where is the function largest?"* only. Ask *"where is the **value of information** largest?"*

Single-objective toy problem

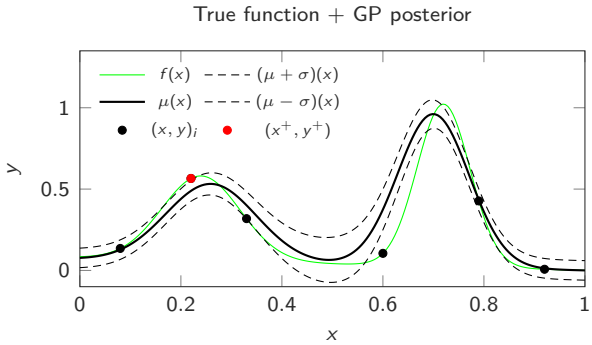
Example

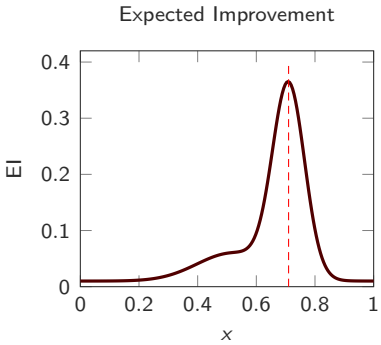
Let $x \in [0, 1]$ be a design variable and suppose the true objective is

$$f(x) = 0.52 \exp\left(-\frac{(x-0.24)^2}{0.012}\right) + \exp\left(-\frac{(x-0.72)^2}{0.0055}\right) - 0.08x + 0.08$$

- We only observe a few samples of $f(x)$
- The GP gives a posterior mean $\mu(x)$ and standard deviation $\sigma(x)$
- Expected Improvement (EI) selects the next experiment

Expected Improvement chooses the next experiment





Takeaway

Exactly like in active learning, we do not pick the point with largest mean only. We pick the point with largest **expected improvement**.

From one objective to two

Two-objective toy problem

Now optimize two objectives at the same time

Suppose one design variable $x \in [0, 1]$ controls two properties:

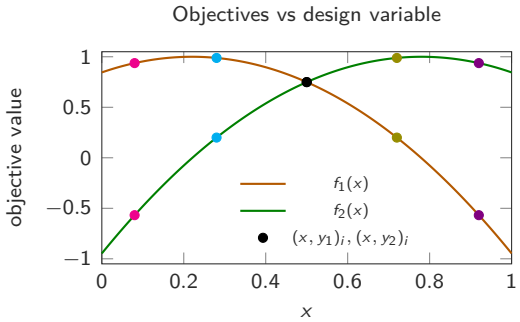
$$f_1(x) = 1.00 - 3.2(x - 0.22)^2, \qquad f_2(x) = 1.00 - 3.2(x - 0.78)^2$$

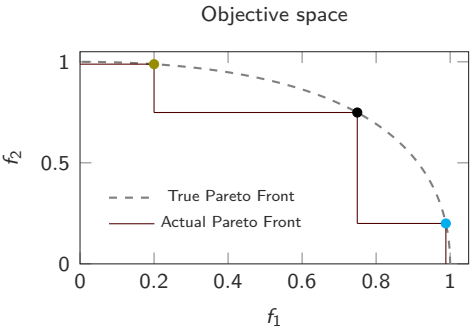
- f_1 prefers designs near $x \approx 0.22$
- f_2 prefers designs near $x \approx 0.78$
- Improving one objective usually hurts the other

Consequence

There is no single best point anymore. Instead, we look for a set of trade-off solutions.

The trade-off is easy to see in objective space

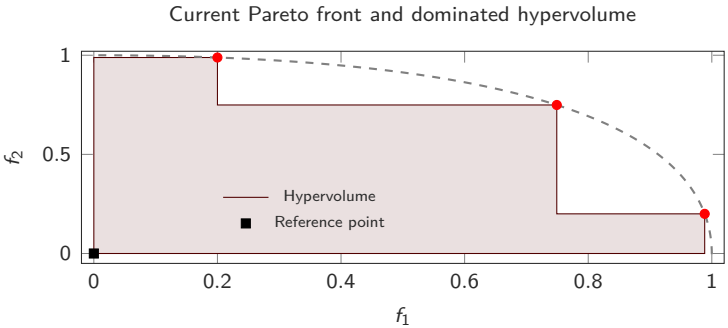




Idea

In multi-objective optimization, the goal is not one optimum value. The goal is to approximate the **Pareto front**.

Pareto front and hypervolume



- **Pareto front:** non-dominated objective values
- **Hypervolume:** volume of objective space dominated by the Pareto front and bounded by a reference point

Key definitions

Definitions

Pareto dominance

For a maximization problem, point $\mathbf{y}^a$ dominates $\mathbf{y}^b$ if

$$y_i^a \geq y_i^b \text{ for all } i,$$

and strictly better in at least one objective.

Pareto front

The Pareto front is the set of non-dominated objective vectors.

Definitions

Hypervolume

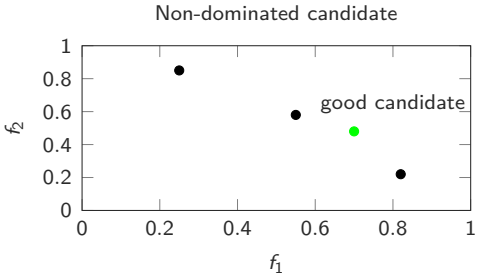
Hypervolume measures how much objective space is covered by the current Pareto front relative to a reference point.

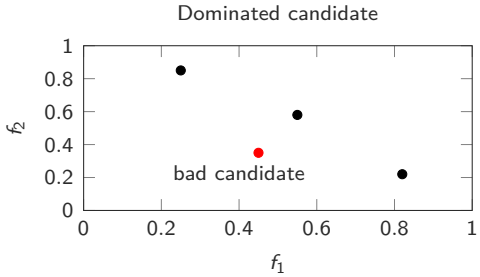
Expected Hypervolume Improvement (EHVI)

EHVI chooses the next experiment with the largest **expected increase in hypervolume**.

$$\mathrm{EHVI}(x) = \mathbb{E}\left[HV(\mathcal{P} \cup \{\mathbf{Y}(x)\}) - HV(\mathcal{P})\right]$$

Simple examples of non-dominated and dominated points





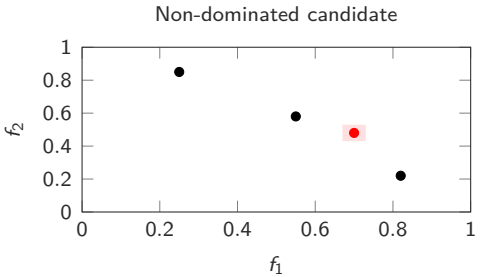
This point matters

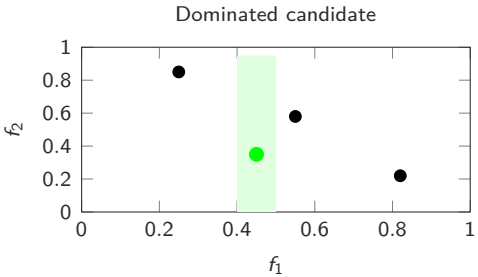
It is not dominated by the current front, so it can improve the Pareto set.

This point does not help

It is already dominated, so it adds essentially no hypervolume.

Remember Surrogate Model Predictions Include Uncertainty





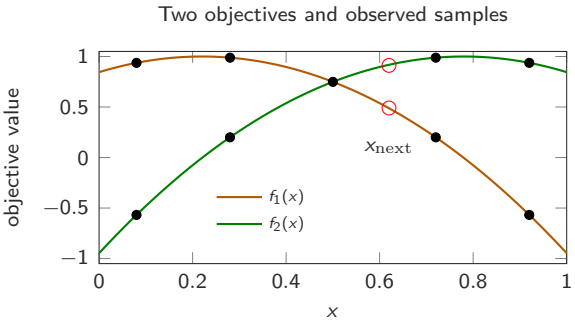
Not necessarily the best

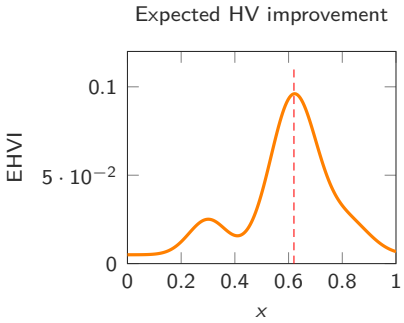
It has descent mean values but uncertainty is low, so EHVI may not be that high.

Can be a good candidate

Mean values are not that good but uncertainty is very large so EHVI can be higher.

Maximizing EHVI chooses the next design





Interpretation

Just like EI in the single-objective case, EHVI is an acquisition function over the input space. We choose the next point by maximizing it.

Three-objective case

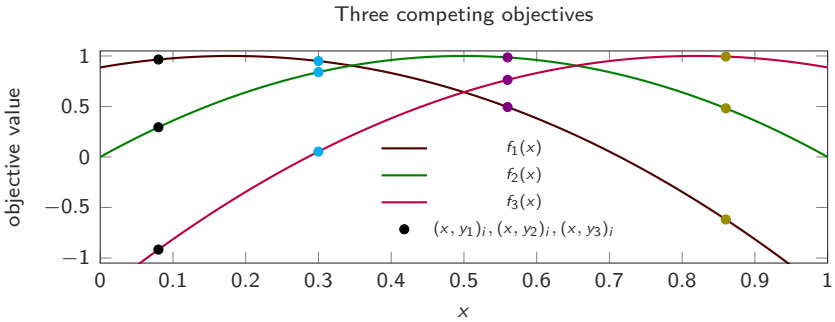
Three-objective toy problem

Now optimize three properties at once

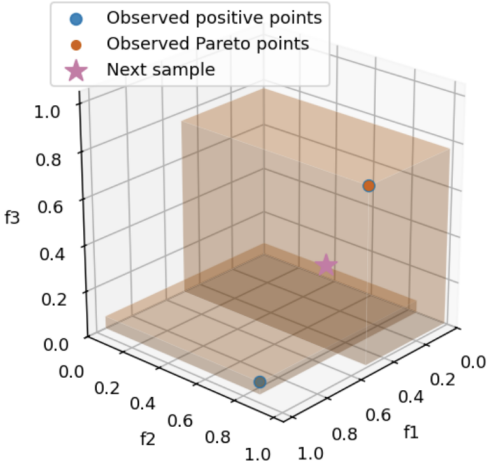
$$f_1(x) = 1.00 - 3.5(x - 0.18)^2,$$
$$f_2(x) = 1.00 - 4.0(x - 0.50)^2,$$
$$f_3(x) = 1.00 - 3.5(x - 0.82)^2$$

- One design variable can no longer maximize all three objectives simultaneously
- The Pareto set becomes a surface or cloud in 3D objective space
- Hypervolume is now a **volume** instead of an area

Objective functions for the three-objective case



Initial Pareto set and hypervolume in 3D objective space



Takeaways

Takeaways

- Single-objective BO chooses the next experiment by maximizing an acquisition function such as EI
- Multi-objective BO replaces "best value" with a **Pareto front**
- Hypervolume gives a practical way to measure the quality of the current Pareto set
- EHVI is the natural multi-objective analogue of EI
- The logic is the same in both cases: choose the next experiment with the largest expected improvement

Thank You!

Questions?